

Lab 06

Sunspot Challenge – Open-Ended Lab

Name:	Student ID:
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6.1 Objective

The aim of this open-ended lab is to analyze and reconstruct sunspot data by applying signal processing and modeling techniques in MATLAB.

6.2 Background

A flare is defined as a sudden, rapid, and intense variation in brightness. A solar flare occurs when magnetic energy that has built up in the solar atmosphere is suddenly released. The amount of energy released is the equivalent of millions of 100-megaton hydrogen bombs exploding at the same time. The first solar flare recorded in astronomical literature was on September 1, 1859. Two scientists, Richard C. Carrington and Richard Hodgson were independently observing sunspots at the time when they viewed a large flare in *white light*.

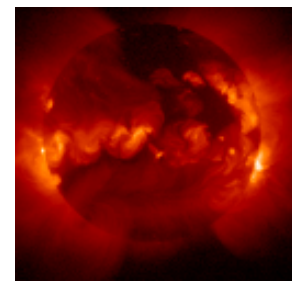


Figure 1: Solar Flares.

Sunspots are areas that appear dark on the surface of the Sun. They appear dark because they are cooler than other parts of the Sun's surface. The temperature of a sunspot is still very hot though—around 6,500 degrees Fahrenheit.

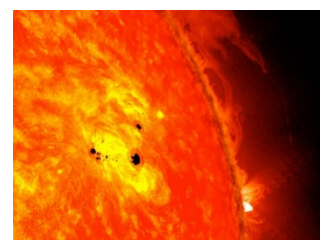


Figure 2: Sunspots

The sun outages create disruption in satellite communications. During sun outages, different interruptions such as sparkles, distorted images, audio distortion, and channel loss are observed. As a result of sun outages, certain satellites may be damaged or knocked out of service.



Figure 3: Solar Flares hitting the satellite



imminent larger solar flares was proposed by the institute for Space-Earth Environmental Research (ISEE), Nagoya University.

6.3 Main goal

The main goal of this lab is to learn DE corruption of inversible and non-inversible distortions and then evaluating the accuracy of the fit by analyzing the power of the fitting error in decibels (dB).

Task 1

The main challenge of this lab is to extract the measured sunspots data year wise from 1700-2025 using MATLAB in .csv and .mat format.

- How did you find the data?
- How was it downloaded?
- How did you convert the data to .mat and .csv format?
- At the end of this Lab create a folder of your name populate it with your .m/.mlx file and Sunspots measured data (in .mat and .csv format) and submit it in the .m/.mlx file submission tab.

Task 2

- Suppose sunspots data got point-wise multiplied by a vector $V=[V_1 V_2 V_3 \dots V_N]$. Plot the **Original data** and the **Corrupted data**. How are they different?

- Sunspot data got multiplied by a matrix $M = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \dots & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{bmatrix}$. Implement this

operation and also show how would you recover the original?

- Now you are going to corrupted the sunspot data by clipping the upper limit of sunspot data to 70 + last digit of your ID, such types of clippings/distortions are non-invertible. What are the possibilities that it is recoverable and how would you try to recover the original signal? Explain the method used.
- Compare the results of part b and c.

Task 3

Model the July 1985 – June 1996 sunspots cycle as $a-b \cos(c(t-d))$.



Vary parameters a, b, c, d to see which fit better. The goal of this task is to analyze the goodness of fit of a model to the sunspot data. Write a MATLAB function that takes the fit and misfits part and evaluates the residual's power or strength to quantify the goodness of fit.

- a) Clearly state the method you used and justify why it is appropriate.
- b) Compute and report the residual power in dB (or any suitable scale).
- c) Comment on how your method captures the goodness of fit.



Assessment Rubric
Open-Ended Lab
Lab 6: Sunspot Challenge

Name:	Student ID:
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Marks distribution:

	Tasks	OE3 Data collection	OE2 Experimental Design	OE6 Results, Analysis & Interpretation	AR4 Report Submission	OE6 Reflection
In-Lab	Task 1	/10			/10	
	Task 2		/10	/15		
	Task 3		/15	/15		/15
Total =100		/10	/35	/30	/10	/15
CLO Mapped			CLO3	CLO3	CLO3	CLO3

CLO	Total Points	Points Obtained
3	100	

For description of different levels of the mapped rubrics, please refer the provided Lab Evaluation Assessment Rubrics and Affective Domain Assessment Rubrics.

**Open-Ended Lab Evaluation Assessment Rubric**

#	Assessment Elements	Level 1: Unsatisfactory	Level 2: Developing	Level 3: Good	Level 4: Exemplary
OE1	Problem Definition	Fails to clearly define the problem or research question.	The problem has been defined to some extent but lacks precision or relevance.	The problem or research question has been defined clearly and contextually	The problem has been defined clearly/ Research has been done with precision and relevance.
OE2	Experimental Design	The proposed design is unsubstantiated or does not satisfy most of the given constraints. The breakdown of system into features is incomplete and still at lower resolution.	The justification for the proposed design is weak, and it only satisfies some constraints. The breakdown of the system is missing some features and resolution is not appropriate in some places.	The proposed design is substantiated, preferably through proper analysis, and satisfies most given constraints. The breakdown of system into features seems complete, but resolution is not appropriate at some places.	The proposed design is substantiated, preferably through proper analysis, and satisfies all given constraints. The breakdown of system into features seems complete and at appropriate resolution, given students' level.
OE3	Data Collection	Measurements are incomplete, inaccurate and imprecise. Observations are incomplete or not included. Symbols, units and significant figures are not included.	Measurements are somewhat inaccurate and imprecise. Observations are incomplete or vague. Major errors are there in the use of symbols, units and significant digits.	Measurements are mostly accurate. Observations are generally complete. Minor errors are present in using symbols, units and significant digits.	Measurements are both accurate and precise. Data collection is systematic. Observations are very thorough and include appropriate symbols, units and significant digits and tasks completed in due time.
OE4	Analysis & Interpretation	Fails to analyze data effectively or draw meaningful interpretations.	The analysis of data is incomplete or important trends have been missed or provided limited interpretations.	The data has been analyzed effectively, by identifying evident trends and drawing reasonable interpretations.	In depth analysis has been made by identifying trends, patterns, and relationships. Insightful interpretations have been drawn.
OE5	Results, Plots and report	Figures/ graphs / tables are not developed or are poorly constructed with erroneous	Figures, graphs and tables are drawn but contain errors. Titles, captions, units are not accurate. Data	All figures, graphs, tables are correctly drawn but contain minor errors or some of the details	Figures / graphs / tables are correctly drawn, and appropriate titles/captions and proper units are



		results. Titles, captions, units are not mentioned. Data is presented in an obscure manner or not presented at all.	presentation is not too clear. Most of the requirements are included in the report but are not well explained.	are missing. Report meets all prescribed requirements.	mentioned. Data presentation is systematic. Report meets all requirements, and it is prepared in original and creative way to engage readers.
OE6	Reflection	Lacks meaningful self-evaluation or reflection on the experiment.	Offers limited self-evaluation without significant insights.	Reflects on the experiment, identifying areas of success and areas for improvement.	Demonstrates critical self-evaluation, identifying strengths, weaknesses, and areas for improvement in the experimental approach.